
Bibliomania

Online Bookstore System

Database Systems Project

Prepared by Group 4:

Name	ID
Ibrahim Ahmed Alkady	8679
Mohamed Hossam Abdelrazik	8571
Mahinour Mohamed Ali	8638
Ziad Ahmed Shabaan	8720

1 Implemented Features

The system fulfills all core and advanced requirements for a digital bookstore. The implementation includes:

- **User Management:** Secure Login and Signup for Customers and specialized Admin accounts.
- **Book Management:** Full management for books including ISBN, title, authors, publishers, and categories such as Science, Art, Religion, History, and Geography.
- **Automated Replenishment:** Triggers to manage stock integrity and automatic order placement from publishers when stock levels fall below a specific threshold.
- **Order Processing:** Real-time shopping cart management and automated stock deduction upon successful credit card checkout.
- **Reporting:** Comprehensive admin dashboards for sales analysis (monthly and daily), tracking the top 5 customers, and identifying the top 10 best-selling books.

2 System Architecture

2.1 Entity Relationship Diagram (ERD)

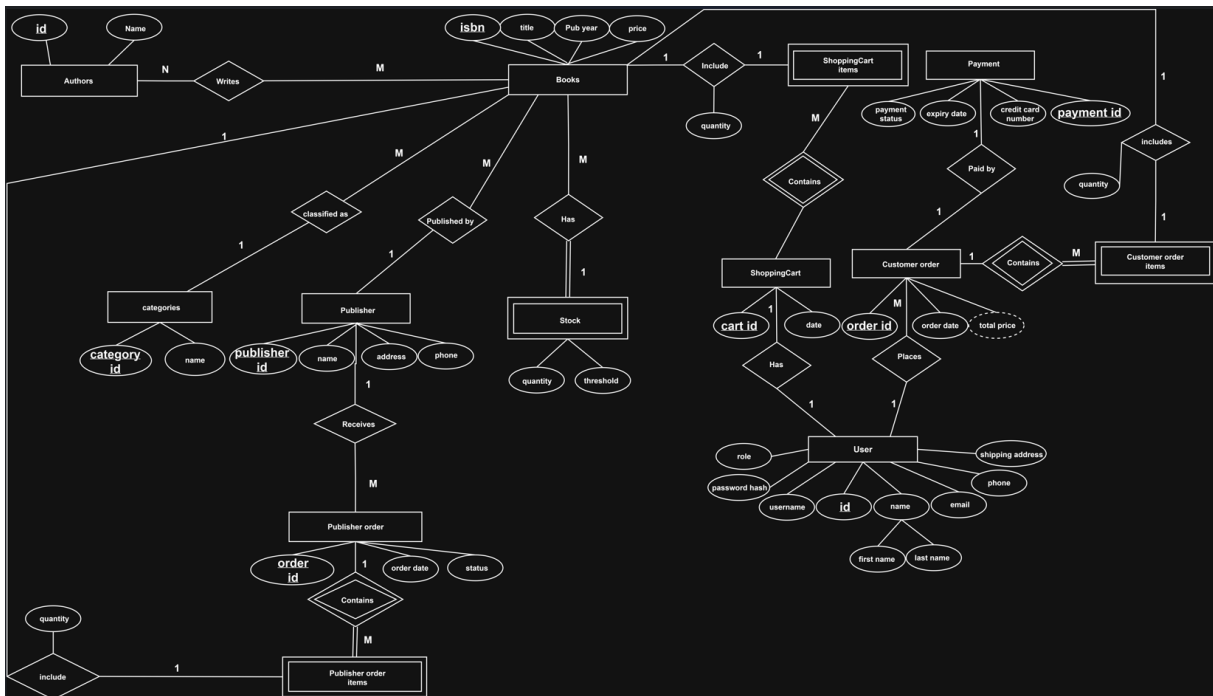


Figure 1: Entity Relationship Diagram for the Bookstore System

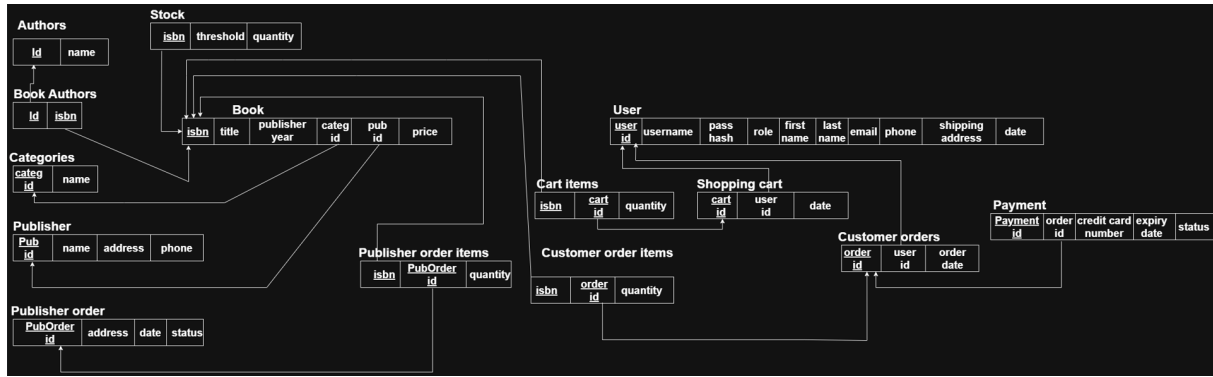


Figure 2: Relational Mapping Schema showing tables and integrity constraints

2.2 Relational Schema

3 User Interface Logic

3.1 Authentication (Login & Signup)

New customers can sign up by providing a username, password, email, and shipping address. The login screen authenticates existing users and determines user privileges, distinguishing between Admin and Customer roles.

3.2 Book Browsing and Search

Users can search for books by ISBN, title, category, author, or publisher. The interface returns comprehensive book details and real-time availability. Admins are granted additional privileges to modify existing book records or add new titles directly from this view.

3.3 Shopping Cart (Slide-out Panel)

Instead of a static page, the cart is a dynamic sliding panel allowing for a seamless user experience.

- **Interactivity:** Users can view total prices and adjust quantities or remove items using a trash icon.
- **Checkout:** Customers provide a credit card number and expiry date. Upon payment validation, the system automatically deducts the purchased quantities from the stock.

3.4 Profile and Order History

Customers can manage their personal information and passwords. A detailed order history section allows users to track past order numbers, dates, purchased items, and total costs.

3.5 Admin Dashboard

A dedicated control center for system management featuring:

- **Analysis Page:** Displays the required sales reports and customer rankings.

- **Publisher Orders:** Provides a dedicated interface to view and confirm replenishment orders to update inventory automatically.
- **Inventory Management:** Real-time tracking of stock levels and threshold-based ordering logic to ensure books remain in stock.